

Generated Summaries as Retrieval Indices for Progressive Retrieval Attention

Compact Addresses and Fixed-Budget Competition

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Abstract

Progressive Retrieval Attention (PRA) separates finding a memory from physically materializing its native keys and values. This separation admits an unusual index: a small language model can summarize each source chunk once, future queries can search those summaries, and a selected address can resolve to unchanged source-derived native K/V. We implement this identity-preserving sidecar and compare frozen generated summaries with lexical, typed extractive, and learned native-QK addresses on identity-disjoint HotpotQA, QASPER, 2WikiMultiHopQA, and MuSiQue cohorts. Standalone summaries help QASPER but tie or regress elsewhere. The follow-up experiment asks the stricter question: does the summary add recall after lexical, extractive, and QK views already share the same four-parent materialization budget? Under validation-frozen reciprocal-rank fusion, adding the summary changes held-out recall by +0.000, −0.113, −0.083, and +0.000, respectively. No positive paired interval excludes zero. Summaries nevertheless recover 2 QASPER and 3 MuSiQue evidence parents missed by every other single channel, so the failure is admission under a fixed budget rather than complete redundancy. Generated summaries remain optional interpretable semantic addresses, not the default index stack. One immutable native-memory record can support multiple compact typed address views, but this pilot closes the LoRA and downstream-generation gates until a selective policy establishes reproducible marginal value.

1 Introduction

Long-context systems face two different problems that are often collapsed into one. The first is *discovery*: deciding which old source chunks might matter to the current query. The second is *consumption*: presenting the corresponding information to the language model in the representation it expects. PRA treats them as separate stages. A small resident index performs discovery; only selected source identities cause native keys and values to enter the active attention set.

This paper asks whether generated natural language can serve as that resident index. Consider a report whose main topic is equipment maintenance but whose appendix records the alias “Northglass.” A fluent summary will probably preserve the main topic. A later query about the alias, however, needs the seemingly minor appendix detail. The summary can be concise and accurate yet still be a poor retrieval address. This is the central tension: ordinary summarization preserves present salience, while routing must preserve unknown future discriminators.

We evaluate that tension without turning the experiment into text retrieval-augmented generation. Each generated address remains bound to an exact source URI, chunk identifier, token span, and source hash. A successful match materializes the original source chunk’s unchanged native K/V; summary text never becomes answer evidence. The implementation therefore tests lossy discovery while preserving native consumption.

The principal finding is mixed but useful. A frozen 0.6B retrieval summary raises held-out QASPER recall from 0.350 to 0.513, but an inherited rank-16 QK selector reaches 0.775. More importantly, the

new multi-index experiment shows that standalone value is not marginal value. A summary can recover a candidate missed by another channel and still reduce final recall if it displaces stronger candidates from a four-slot palette. Controlled low-salience targets explain the pressure: generated addresses often omit the exact entity, alias, date, or rare string that makes a chunk reachable. A cheap extractive index preserves these addresses more reliably.

This study contributes five concrete results:

1. an identity-safe summary sidecar whose output cannot leak into native-K/V materialization;
2. a matched-budget comparison across four natural datasets and inherited lexical and learned-QK baselines;
3. controlled omission and equal-token facet experiments that isolate coverage from fluency; and
4. a fixed-budget multi-index comparison of round-robin, reserved-slot, agreement, reciprocal-rank, and normalized-score admission; and
5. measured index, ingestion, routing, and amortization costs that prevent compact storage from being confused with cheap operation.

2 Where summary indices fit in PRA

Earlier work in the PRA series narrows the question. Paper 2.5 found that iterative associative closure can expose controlled bridges but does not reliably improve natural multihop retrieval. Paper 2.6 found lexical and semantic channels to be complementary. Paper 2.7 showed that query faceting is not automatically beneficial. Paper 2.8 found that learned low-rank native-QK routing can improve selected semantic cohorts while BM25 remains strong for lexical evidence. Paper 2.9 found no reliable held-out gain from short temporal query pooling. Paper 3 then separated conceptual selection from physical native-K/V materialization.

Paper 3.1 changes only the representation used for discovery. It does not compress active K/V, rewrite source state, or answer from summaries. This distinction also separates the method from conventional text RAG: retrieved language is an address that resolves to model-native evidence, not the evidence placed in the prompt.

The result is best understood as a typed-index question. Source BM25 preserves explicit strings. A generated summary can normalize paraphrases and relations. Learned QK features preserve host-model geometry. These representations fail differently, so the scientific question is not whether one replaces all others, but whether generated language adds enough complementary discovery to justify its ingestion cost.

3 Method

3.1 An address backed by native evidence

Let source chunk C_i occupy token span $[a_i, b_i)$ within source URI u_i . Ingestion generates a bounded address

$$S_i = f_\theta(C_i),$$

where f_θ is frozen in the main study. The persistent record is

$$R_i = (u_i, i, a_i, b_i, h(C_i), S_i),$$

where $h(C_i)$ is a SHA-256 digest of the exact source text. The first five fields form the native-memory identity. The final field is replaceable routing metadata.

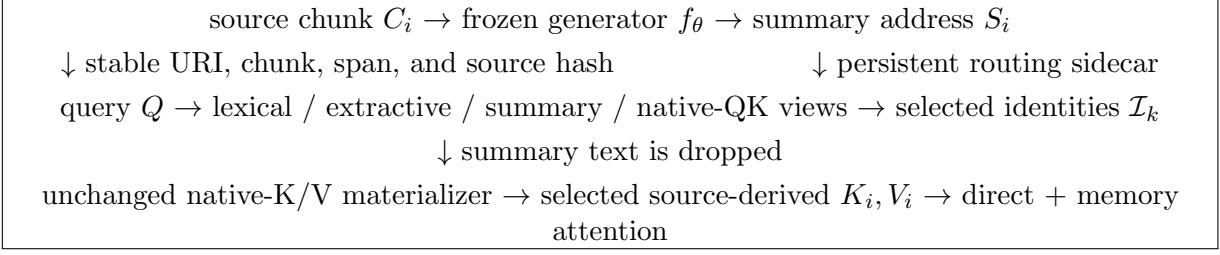


Figure 1: The summary is a lossy address. Selection resolves back to an immutable source identity before native keys and values are materialized.

For query Q , a scorer g assigns one score per record,

$$r_i = g(Q, S_i).$$

The router chooses the top k identities $\mathcal{I}_k$. Materialization then discards S_i and requests

$$\mathcal{M}(Q) = \{(u_i, i, a_i, b_i, h(C_i)) : i \in \mathcal{I}_k\}.$$

The unchanged materializer resolves those identities to source-derived native K/V. This projection is intentionally one-way: the native-K/V request type has no summary field.

3.2 Persistent sidecar and cache safety

The implementation lives in `src/prahf/summary_index.py`. `SummaryIndexRecord` stores the immutable identity beside generated summary text or independently scored facets. `native_kv_request()` removes all lossy text. `assert_source_alignment()` compares the full URI, span, and hash set with the inherited native-memory registry before evaluation.

Generation is deterministic and resumable. The cache key hashes the model, prompt version, token budget, facet count, and exact source text. The cache deliberately excludes the logical item identifier so identical chunks may reuse generated content. A cache hit is rebound to the requesting item identity; this detail matters because content reuse must never alias two native-memory addresses. A regression test covers duplicate-content rebinding.

3.3 Scorer are separate from summaries

Summary quality and query matching are independent variables. We therefore score the same generated address with four frozen channels:

- exact normalized query-token overlap;
- BM25 over summary text;
- cosine similarity from frozen MiniLM embeddings; and
- a validation-frozen convex fusion of BM25 and embedding scores.

For a faceted record, each facet is scored separately and the chunk receives its maximum facet score. The hybrid first min-max normalizes each channel within the source, then computes

$$r_i^{\text{hyb}} = \alpha \tilde{r}_i^{\text{BM25}} + (1 - \alpha) \tilde{r}_i^{\text{emb}}.$$

Weights $\alpha \in \{0.25, 0.50, 0.75\}$ are considered only on validation identities. The held-out policy is never changed after observing test results.

3.4 One backing memory, multiple address views

The next iteration gives every parent identity four aligned address views: source BM25 (L), a deterministic entity/rare-term sidecar (E), the frozen generated summary policy (S), and the five-seed rank-16 native-QK selector (QK). The typed extractive record retains separately detected entities, optional aliases, numbers and dates, corpus-rare terms, high-IDF bigrams, and relation terms. Its extraction is deterministic but not claimed to be semantically lossless.

Each channel produces a score and rank for the same parent i . The router may combine these addresses, but the final set remains bounded:

$$\mathcal{I}_K = \text{Admit}_K(r^L, r^E, r^S, r^{QK}), \quad |\mathcal{I}_K| \leq K.$$

We test rank round-robin, fixed slot reservations, agreement priority, reciprocal rank fusion (RRF), and min-max normalized score fusion. RRF uses

$$r_i^{\text{RRF}} = \sum_{c \in \mathcal{C}} \frac{1}{d + \text{rank}_c(i)}.$$

The small choice $d \in \{10, 60\}$ and five interpretable score-weight templates are frozen separately by dataset and channel combination on validation identities. Every test row retains each selected parent’s score and rank from every participating channel. Sidecar representations are then discarded, and the selected integer parent identities enter the unchanged native-K/V request path.

4 Experimental design

4.1 Data and inherited baselines

We replay identity-disjoint HotpotQA, QASPER, 2WikiMultiHopQA, and MuSiQue examples from Paper 2.8. Source text is retokenized with the pinned Qwen3-0.6B tokenizer, then asserted against inherited 256-token parent spans and evidence masks. Every query selects at most four parent chunks. The materialization budget is therefore matched even when index representations differ.

The baselines are: one native mean key per parent chunk; BM25 over full source text; an inherited five-seed rank-16 native-QK selector; an inherited five-seed compact rank-8/eight-centroid selector; and oracle identity scores. The oracle is a ceiling on discovery, not an available system.

The generated-model frontier includes 135M, 0.6B, 1.2B, 4B, and 8B local models. Small frontier runs establish feasibility only. A two-identity-per-dataset validation run selects one model, prompt, and scorer per dataset. The held-out cohort then contains four HotpotQA, eight QASPER, four 2Wiki, and eight MuSiQue identities. This is a pilot cohort: paired bootstrap intervals quantify identity uncertainty but do not make it adequately powered.

4.2 Frozen policies

Validation selected four different policies, itself evidence against a universal summary channel. HotpotQA uses an 8B generic summary with exact scoring. QASPER uses a 0.6B retrieval summary with exact scoring. 2Wiki uses an 8B retrieval summary with BM25. MuSiQue uses a 0.6B retrieval summary with equal BM25–MiniLM fusion. All summaries have an upper bound of 32 generated tokens per source chunk.

4.3 Fixed-budget multi-index protocol

The follow-up reuses exactly those generated addresses; no summary is regenerated or selected from test behavior. Eight validation identities, two per dataset, freeze RRF constants and normalized-fusion

templates. The held-out cohort is the same 24 identities used above. We sweep $K \in \{2, 4, 8\}$ and treat $K = 4$ as primary. Candidate pools may be larger internally, but every policy admits at most K unique source parents.

Single-channel baselines retain native mean, source BM25, typed extractive, generated summary, rank-16 QK, rank-8/eight-centroid QK, and oracle identity. The multi-index matrix includes every pair, the requested three-channel stacks, and $L + E + S + QK$. Round-robin and agreement union are parameter-free. Reserved policies include $2L + 2S$, $2E + 2S$, $2L + 2QK$, $2E + 2QK$, and interpretable one- or two-slot lexical safety reservations. Score fusion searches only equal, address-weighted, semantic-weighted, summary-weighted, and QK-weighted templates on validation. RRF searches only $d = 10$ or 60. This is deliberately not a combinatorial controller search.

4.4 Metrics and gates

The primary endpoint is evidence recall at four selected parent chunks. We also report complete recovery, precision, reciprocal rank, persistent index bytes, summary tokens, ingestion time, routing time, and native-K/V tokens requested. Bootstrap intervals use 10,000 paired identity resamples. Multi-index comparisons use identities, not channels or evidence parents, as the paired statistical unit. Address-retention correlations are descriptive diagnostics over positive parents and are not treated as independently powered claims.

Training is gated. LoRA distillation begins only if frozen summaries show a positive held-out signal or if a teacher establishes clear headroom over a small model. Native-K/V answer generation begins only after a discovery condition passes the retrieval gate. These gates prevent capacity or downstream adaptation from obscuring a failed index hypothesis.

5 Held-out retrieval results

Figure 2 and Table 1 report the frozen policies. QASPER provides the only positive summary delta: recall rises from 0.350 to 0.513. The paired interval is broad, $[-0.400, 0.688]$, and rank-16 QK reaches 0.775. The summary is therefore a promising complementary QASPER address, not the best standalone index.

HotpotQA does not confirm the teacher’s validation advantage. The summary and native mean both reach 0.875, while source BM25 reaches 1.000. On 2Wiki, the teacher summary falls to 0.708 from 0.917 for native mean, source BM25, and both QK selectors. On MuSiQue, the frozen hybrid reaches 0.563 versus 0.615 for native mean and 0.646 for source BM25. None of the four paired intervals excludes zero in the positive direction.

Dataset	n	Mean	Src. BM25	R16	R8/C8	Summary	Oracle	Δ [95% CI]
HotpotQA	4	.875	1.000	.875	.750	.875	1.000	.000 [-.375,.375]
QASPER	8	.350	.325	.775	.700	.513	.925	+.163 [-.400,.688]
2Wiki	4	.917	.917	.917	.917	.708	1.000	-.208 [-.417,.000]
MuSiQue	8	.615	.646	.344	.458	.563	1.000	-.052 [-.375,.208]

Table 1: Evidence recall at matched selected-chunk budget. Δ is the paired summary-minus-native-mean effect. R16 and R8/C8 are inherited learned native-QK selectors.

Complete recovery follows the same pattern. The summary policies recover every evidence chunk on 0.750, 0.375, 0.500, and 0.250 of HotpotQA, QASPER, 2Wiki, and MuSiQue examples. Thus the QASPER recall gain is not merely a partial-hit artifact, but it also does not close the oracle gap.

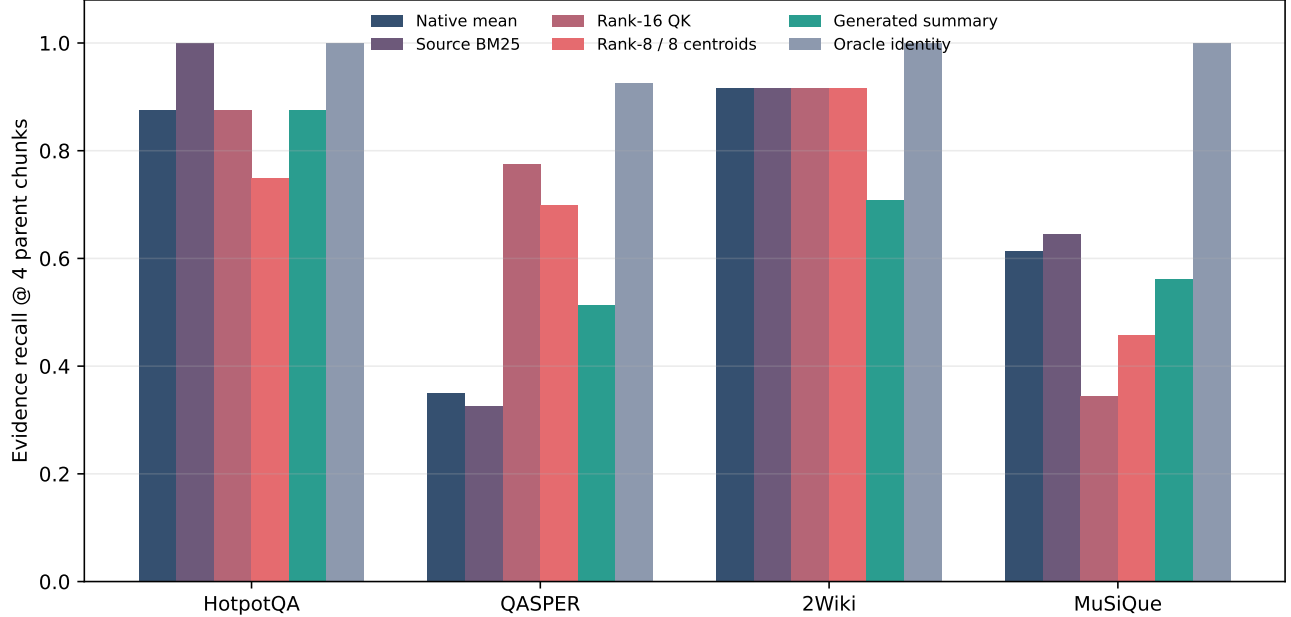


Figure 2: Held-out evidence recall at four selected parent chunks. “Generated summary” denotes the scorer and generator frozen independently for each dataset on validation identities.

6 Fixed-budget multi-index results

6.1 The channels are complementary

Figure 3 classifies each evidence parent by which single-channel top-four lists recover it. All eight HotpotQA evidence parents are multi-channel recoveries. QASPER is less redundant: QK alone recovers four evidence parents, S alone recovers 2, L alone recovers one, and six are missed by every channel. MuSiQue has 3 summary-only, two lexical-only, and one extractive-only recovery. Summaries therefore contain unique routing signal; the remaining question is whether that signal survives competition for four slots.

6.2 More address views do not imply more admitted evidence

Table 2 reports the validation-frozen RRF comparison at the same $K = 4$ materialization budget. $L + S$ matches or exceeds either member on HotpotQA and MuSiQue, and $L + QK$ reaches 1.000 on 2Wiki. The full four-view stack does not dominate. It ties the $L + E + QK$ base on HotpotQA and MuSiQue, falls on QASPER, and loses one 2Wiki evidence outcome. The cost frontier in Figure 4 makes the architectural accounting explicit: these are multiple resident address views over one backing memory, not multiple copies of native K/V .

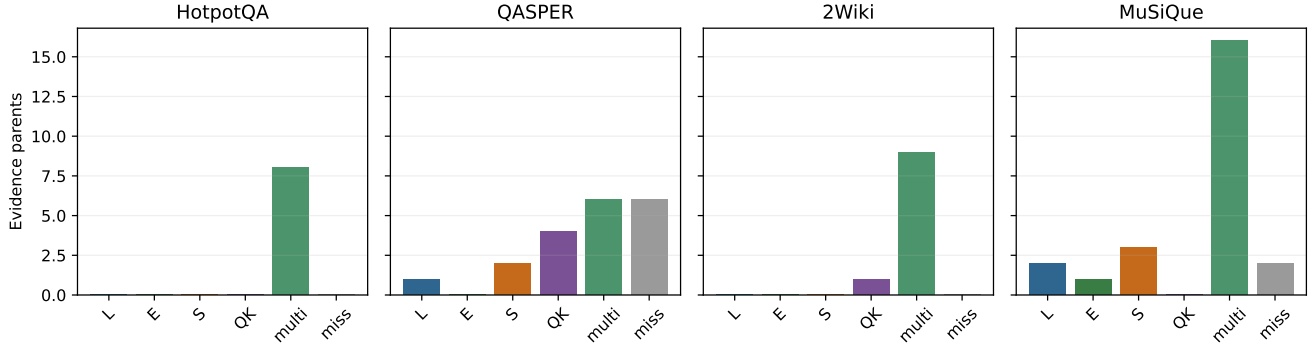


Figure 3: Evidence-parent recovery by single-channel top-four lists. “Multi” means at least two of L, E, S, QK recover the parent; “miss” means none does. Counts describe complementarity before a combined admission policy.

Policy	HotpotQA	QASPER	2Wiki	MuSiQue
L	1.000	0.325	0.917	0.646
E	0.875	0.138	0.917	0.552
S	0.875	0.512	0.708	0.562
QK	0.875	0.775	0.917	0.344
L+S (RRF)	1.000	0.475	0.917	0.688
L+QK (RRF)	1.000	0.588	1.000	0.542
L+E+QK (RRF)	1.000	0.588	1.000	0.573
L+E+S+QK (RRF)	1.000	0.475	0.917	0.573

Table 2: Held-out evidence recall at $K = 4$. Single-channel rows use their own rankings. Combined rows use validation-frozen RRF and the same final parent budget.

6.3 Summary marginal value depends on admission

The primary marginal test is

$$\Delta_S = R(L + E + S + QK) - R(L + E + QK).$$

Under RRF, Table 3 gives deltas of +0.000, −0.113, −0.083, and +0.000. No positive interval excludes zero. QASPER has one win, six ties, and one loss; the negative mean comes from the larger loss. On 2Wiki, one loss and three ties produce the −0.083 delta.

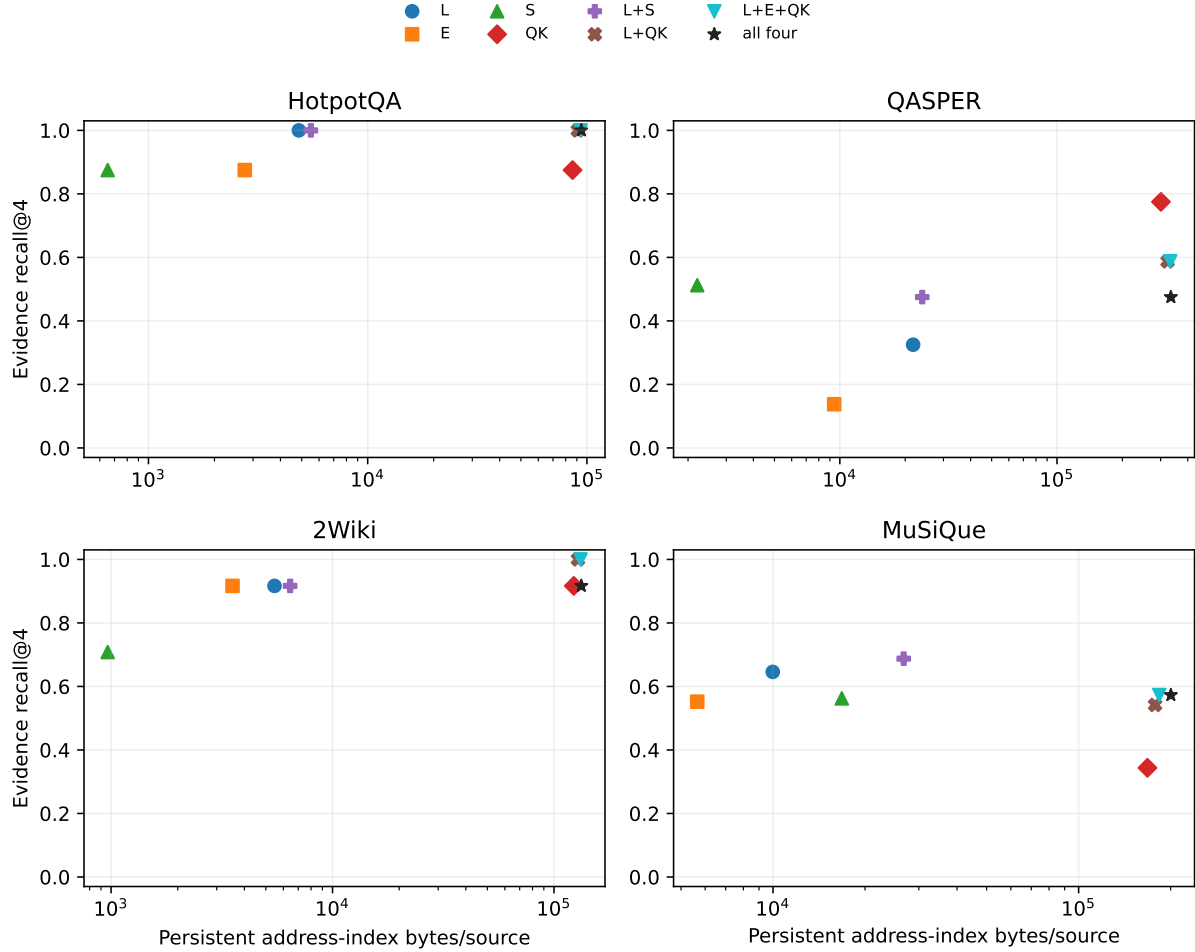


Figure 4: Fixed-budget recall versus persistent address-index bytes per source. Backing source/native-K/V storage is shared and excluded from every point.

Dataset	Base	+Summary	Δ_S	95% CI
HotpotQA	1.000	1.000	+0.000	[+0.000, +0.000]
QASPER	0.588	0.475	-0.113	[-0.375, +0.037]
2Wiki	1.000	0.917	-0.083	[-0.250, +0.000]
MuSiQue	0.573	0.573	+0.000	[+0.000, +0.000]

Table 3: Summary marginal recall under validation-frozen RRF. “Base” is $L + E + QK$; “+Summary” is $L + E + S + QK$. Intervals use 10,000 paired identity resamples.

Other admission rules reveal small but unresolved positive regimes. Round-robin changes QASPER by +0.013 [+0.000, +0.038] and MuSiQue by +0.073 [+0.000, +0.167]. A fixed $2L + 2S$ MuSiQue reservation reaches 0.719, a +0.073 effect versus L alone, but its interval $[-0.115, +0.260]$ also crosses zero. Agreement priority and normalized fusion likewise fail to produce a resolved positive summary marginal. The policy-level artifact reports paired effects against every constituent single channel, so no favorable point is promoted without its interval and wins/ties/losses.

The distinction matters. Summaries are *complementary*: they find evidence missed by L , E , and QK . They do not yet show *net fixed-budget utility* because unconditional admission can displace stronger candidates. Selective summary admission remains plausible; a default four-view union does not.

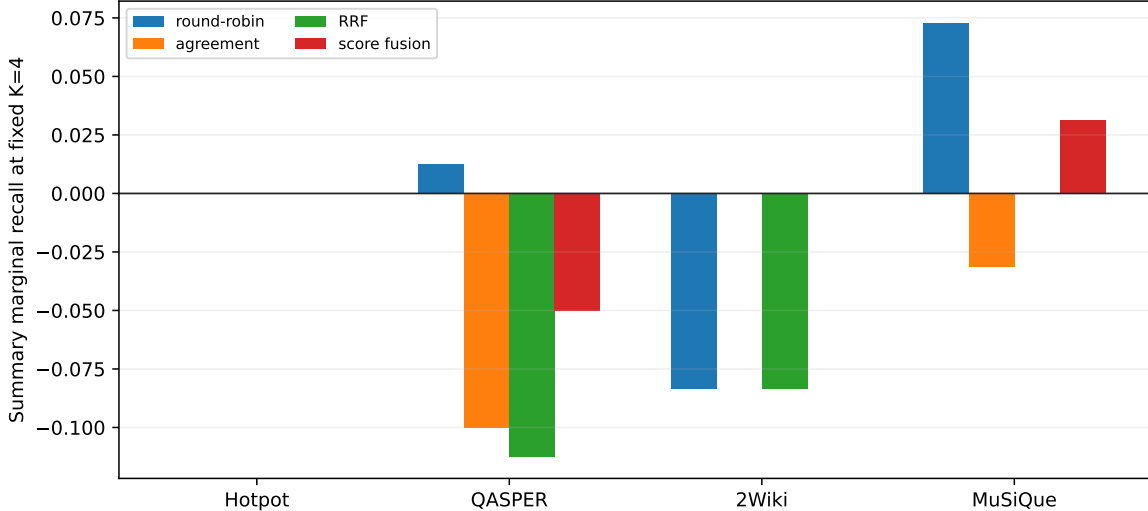


Figure 5: Summary marginal recall across four fixed admission families. All comparisons hold $K = 4$ constant.

7 Why summaries fail: controlled omission

Natural retrieval cannot reveal whether a chunk was missed because the generator omitted an address or because the scorer failed to use one. We therefore construct 20 examples, four per fact type. Each example contains six topical chunks. One chunk hides a query-critical entity, alias, relation, date/number, or rare string as a minor detail. The target does not influence summary generation.

The full-source BM25 and the 32-token entity/rare-term extractive index recover every target chunk. The 0.6B retrieval summary also recovers all relation, date/number, and rare-string chunks, but only 0.75 of entity chunks and 0.25 of alias chunks. Crucially, it retains no complete target literal in any fact category. Retrieval sometimes succeeds because other generated wording correlates with the positive chunk; the shuffled-address control confirms that small-cohort ties and shared topical language can also produce nonzero recall.

The failure is conceptual rather than merely linguistic. A summary can be faithful while deleting the only future retrieval key. Retrieval-oriented prompting helps preserve classes of information, but a fixed 32-token language channel cannot guarantee coverage of every unknown discriminator. This supports a typed design: retain a lexical or entity sidecar for exact reachability and use summaries for paraphrastic or relational access.

7.1 Natural address-preservation diagnostics

We apply deterministic diagnostics to the held-out positive source parents. Regex entities, numbers/dates, source-local rare terms, relation terms, and query-overlapping evidence keys are measured against the frozen summary text. Aliases are omitted because these natural cohorts do not provide independent chunk-level alias labels. Figure 7 shows substantial dataset heterogeneity and generally low literal preservation. Across available positive parents, summaries retain only 0.276 of query-overlapping evidence keys.

Evidence-key retention is the strongest measured descriptive predictor of summary recovery, with pooled Spearman $\rho = 0.442$ for target recovery at four. The unit here is a positive source parent, so this is a diagnostic association, not an independently powered inferential result. Figure 8 also shows that length and compression ratio correlate with recovery, consistent with a capacity-coverage tradeoff rather

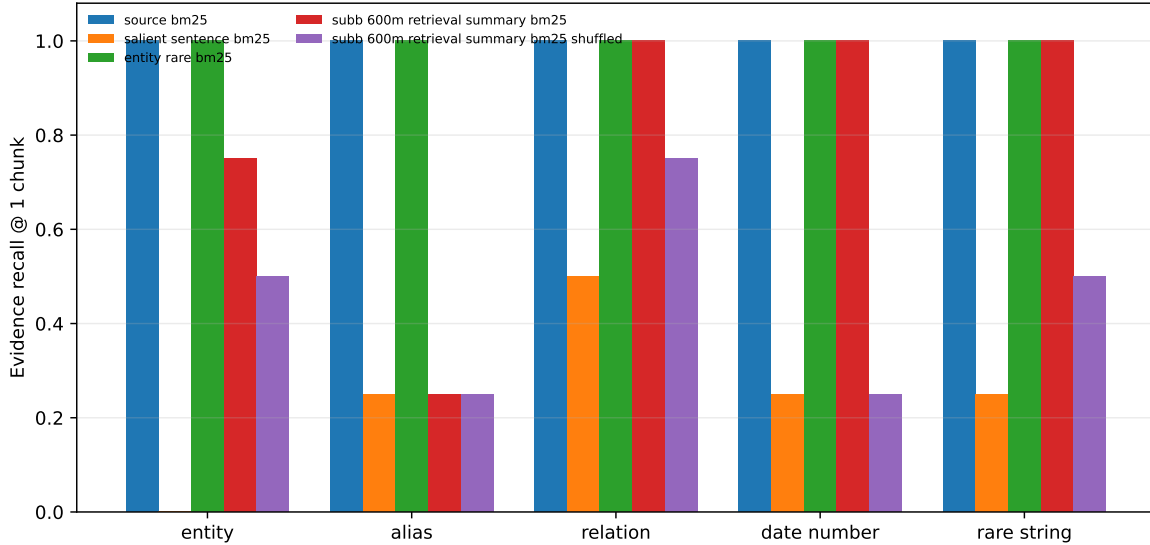


Figure 6: Controlled low-salience fact retrieval at one selected chunk. A parameter-free entity/rare-term sidecar is both smaller and more reliable than generated summaries on this literal-address task.

than a generic fluency deficit.

We do not report ROUGE, BLEU, or BERTScore. None of the four datasets supplies an independently authored chunk-level retrieval-summary reference for these records. Comparing a generated address against its full source and calling the result conventional summary quality would mostly remeasure lexical coverage and compression. The direct address-retention and retrieval correlations are better aligned with the question tested here.

8 Equal-budget facet geometry

Facets might spread a fixed budget across more independent addresses. We compare 1×32 , 2×16 , 4×8 , and 8×4 maximum-token geometries with a frozen 1.2B model on two validation Hotpot identities. BM25 recall is 0.50, 0.75, 0.50, and 0.50, respectively. The small diagnostic favors two medium facets, not monotonically more facets.

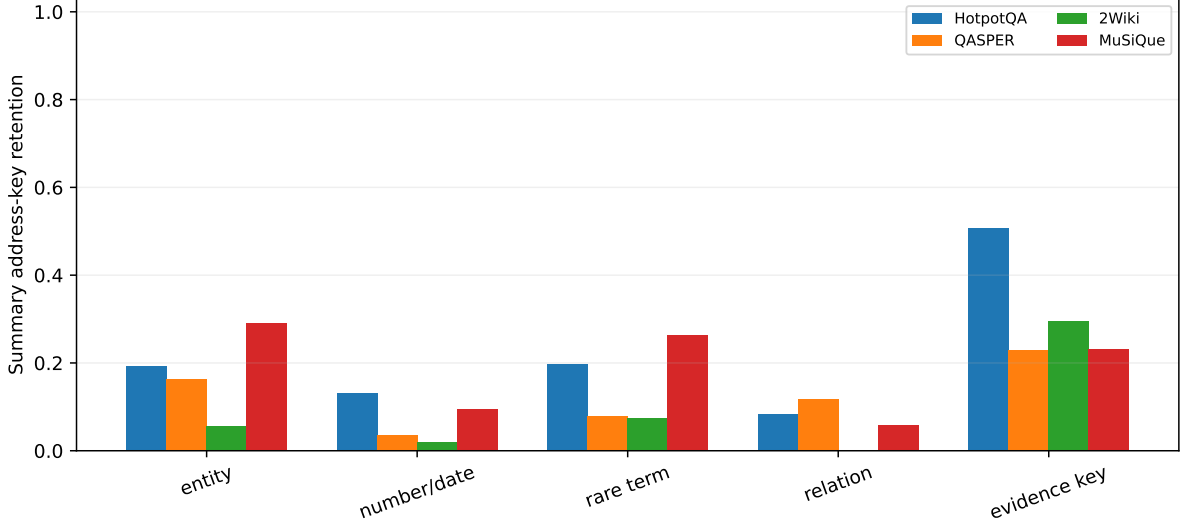


Figure 7: Deterministic address-key retention in frozen summaries of positive source parents. Missing reference categories are excluded rather than scored as zero.

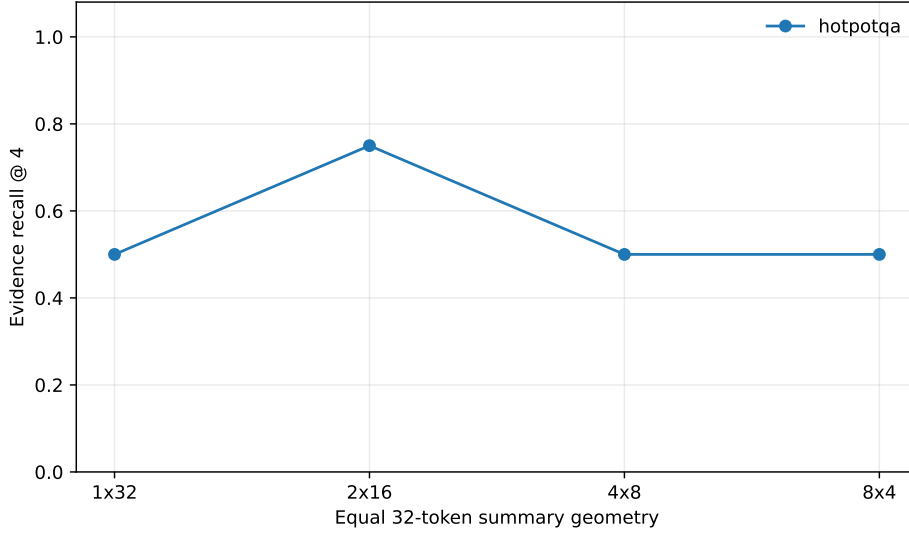


Figure 9: Diagnostic equal-budget facet geometry on two HotpotQA validation identities. The experiment is intentionally not a held-out cross-dataset claim.

Actual output is shorter than the maximum: the four geometries use 184, 172, 151, and 152 summary tokens per source across all candidate chunks. Ingestion takes 33.7, 33.4, 47.9, and 48.9 seconds per source. Structured output overhead therefore makes the many-facet conditions more expensive even when content-token capacity is held fixed.

9 Storage and time accounting

Table 4 separates the persistent routing index from backing source/native-K/V storage and from the physically materialized native K/V. Pure lexical summary policies occupy only 652–2,205 bytes per source. MuSiQue’s selected hybrid occupies 16,793 bytes because 15,744 bytes are frozen embedding vectors;

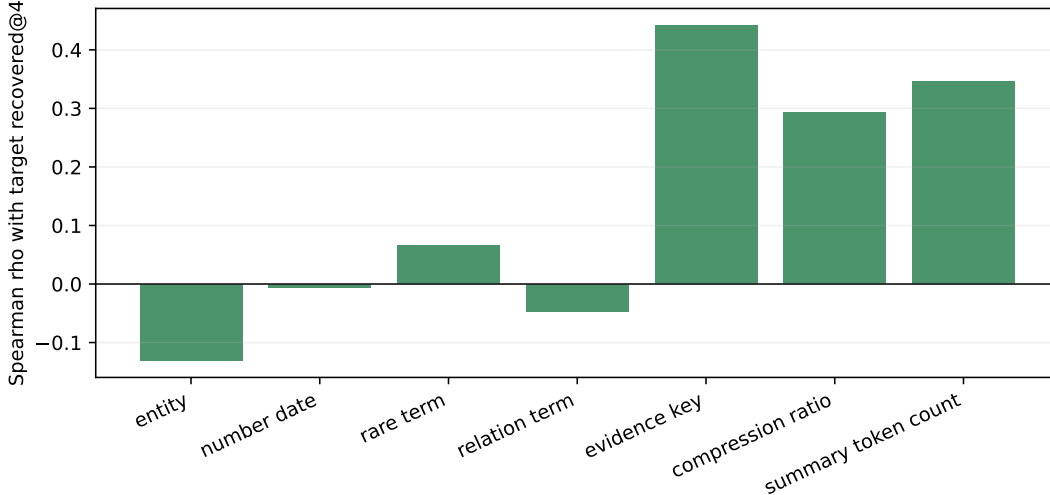


Figure 8: Pooled descriptive correlation between address diagnostics and summary target recovery. Positive values mean higher diagnostic values tend to coincide with recovery.

summary text itself occupies about 1,049 bytes. None of these numbers includes backing native memory.

Dataset	Index B	Emb. B	Tokens	Ingest s	Route ms	Native tokens
HotpotQA	652	0	166	131.1	0.186	1020
QASPER	2205	0	483	46.6	0.561	1020
2Wiki	966	0	240	176.8	0.453	904
MuSiQue	16793	15744	280	25.3	371.4	903

Table 4: Mean per-source accounting for each frozen summary policy. “Index B” includes embedding bytes when the selected scorer requires them. Native tokens are materialized after routing and remain governed by the common four-parent budget.

Table 5 reports the combined resident indices. The base stack stores lexical text, the typed extractive sidecar, and rank-16 QK once. Adding S charges only summary text and, for MuSiQue, its frozen embedding. The shared backing source/native-K/V record is counted once in both conditions. Summary generation dominates one-time ingestion even when its persistent byte increment is small. QK projection/index construction is inherited from the frozen Paper 2.8 artifacts and is marked as not remeasured rather than assigned a zero-cost claim.

Dataset	Base bytes	+Summary bytes	Summary ingestion (s)
HotpotQA	93621	94273	131.1
QASPER	332221	334426	46.6
2Wiki	131886	132852	176.8
MuSiQue	183568	200361	25.7

Table 5: Mean persistent address bytes and measured summary ingestion time per source. “Base” is $L + E + QK$ and “+Summary” is $L + E + S + QK$. Both use one shared native-memory backing record.

For Q later queries to one source, measured one-time generation C_s contributes C_s/Q seconds per query. At $Q = 1000$, that term is 0.131, 0.047, 0.177, and 0.025 seconds for HotpotQA, QASPER, 2Wiki, and MuSiQue. Considering routing time alone, generation amortizes against native-mean scoring

after approximately 479,000, 25,800, and 7,300 queries on the first three datasets. MuSiQue’s external embedding route is slower and never breaks even in this local implementation. Against source BM25, the corresponding thresholds are about 70,000, 9,800, 171,000, and never.

These are not favorable end-to-end break-even claims. The conditions with lower routing cost do not also provide a reliable quality gain. The measurements instead establish the scale at which asynchronous ingestion could become acceptable for repeatedly queried, stable corpora.

10 Gate decisions and implications

Frozen concept gate. QASPER shows standalone summary value, and channel-overlap rows confirm unique summary recoveries on QASPER and MuSiQue. The stricter marginal gate remains closed: adding S to $L + E + QK$ produces no resolved positive effect under RRF, round-robin, agreement, or normalized fusion. We therefore classify summaries as optional candidate sources rather than members of the default resident stack.

Teacher and training gate. The 8B teacher does not establish general headroom: it ties on held-out HotpotQA and regresses on 2Wiki. More importantly, the frozen summary has no reliable marginal value after stronger companion views are present. Training a LoRA summarizer would consequently optimize into a weak and dataset-specific target. No LoRA distillation is run in this iteration.

Native-use gate. The identity projection, requested token counts, and unchanged materialization policy are verified, but no answer-likelihood or generation claim is made. The multi-index discovery gate is not strong enough to justify a downstream native-K/V study. Summary-only answering is also omitted because it would test a different text-RAG system after the relevant gate closed.

Architectural implication. The supported design is one immutable native-memory record with multiple typed address views. Exact tokens, aliases, and identifiers remain with lexical or extractive indices. Learned native-QK channels cover host-model geometry. Generated summaries contribute inspectable relational language only when a selective policy can justify a slot. The router deduplicates identities before one common native-K/V materializer. The multi-index result replaces a vague “hybrid” recommendation with an explicit admission requirement: channel complementarity is useful only when final-slot competition is calibrated.

11 Limitations

The held-out cohorts contain only 24 identities, and two teacher datasets contain four identities each. Bootstrap intervals appropriately remain wide. Multi-index fusion choices use only two validation identities per dataset; the tiny grid limits overfitting but cannot make policy selection stable. The equal-budget geometry study is a two-identity validation diagnostic. The omission benchmark is controlled and intentionally adversarial; its absolute rates should not be generalized to natural corpora.

Generation uses local quantized Ollama models and one machine with a 4GB GPU. The 8B teacher is CPU/GPU split, so its latency should not be treated as a hardware-independent model benchmark. Peak ingestion memory was not sampled robustly; this paper reports wall time, persistent bytes, and model metadata instead. MiniLM timing includes local service overhead and is not an optimized in-process embedding implementation.

The study measures discovery, not final answer quality. Although selected records map to unchanged native-memory requests, no downstream generation is run after the gate closes. Natural alias labels and independent reference summaries are unavailable, so alias retention and conventional ROUGE-style quality are not reported. The frozen cache also caps output at 32 tokens; extending it to 64 or 128 would

require a new generation intervention, not a post-hoc truncation comparison. Finally, a stronger learned summarizer might outperform frozen prompts, but the absent teacher and multi-index headroom provides no evidence that scaling this particular target is the right next step.

12 Reproducibility

Every row is tracked with dataset identity, selected and positive parent indices, index bytes, generated tokens, ingestion and routing time, and materialized native-K/V tokens. Manifests record source-model revision, Ollama model metadata, prompt version, generation and dataset seeds, exact hashes of inherited feature tensors, and the source commits for Papers 2.5 through 3. Generated addresses are append-only JSONL records; source text and large native-QK tensors are not duplicated into the repository.

The original runner is `experiments/paper3_1_summary_index/run_study.py`. The multi-index replay is `run_multi_index_study.py`; `build_multi_index_artifacts.py` regenerates paired effects, tables, macros, and five figures. Required artifacts retain per-candidate scores and ranks, channel overlap, every union/fusion row, typed-address coverage, correlations, component costs, and validation-frozen policy choices. Focused tests verify source alignment, typed extraction, deterministic unions, reserved slots, RRF, normalized fusion, native-request projection, cache resumption, and duplicate-content identity rebinding. The paper’s numerical tables and prose macros are generated from tracked row-level CSVs.

13 Conclusion

Natural-language summaries can be compact, inspectable routing addresses for native memory. They are not reliable universal addresses. In this pilot, one QASPER policy improves over a native mean key, and summaries uniquely recover several QASPER and MuSiQue evidence parents. Yet those unique candidates do not yield reliable marginal recall after lexical, typed extractive, and learned-QK views compete for the same four slots. Controlled omission and natural address diagnostics show the deeper reason: fluent compression can remove exactly the low-salience string that a future query needs.

The useful outcome is architectural clarity: one immutable native-memory record can support several compact address views. Lexical and extractive views preserve explicit reachability; QK preserves host-model geometry; summaries provide a lossy, human-readable semantic address. Their scores must be combined by a fixed-budget admission policy before selected identities resolve to unchanged native K/V. The default stack remains $L + E + QK$. Summary routing is an optional experimental channel until larger cohorts demonstrate positive marginal value; only then should learned summarization or downstream generation reopen.